

# Mostafa Moghaddas

## Curriculum Vitae

[m\\_moghaddas@alumni.iust.ac.ir](mailto:m_moghaddas@alumni.iust.ac.ir)  
Google Scholar  | LinkedIn  | GitHub 

### Research Interests

Hardware Security ■ Cryptographic Engineering ■ Side-Channel & Fault Analysis ■ Physical Design & SoC Integration

### Education

- Oct 2021 – Apr 2024 **M.Sc in Computer Architecture Engineering**, *Iran University of Science and Technology (IUST)*, Tehran, Iran.  
*Thesis*: Providing a Physical Unclonable Function to Improve Resilience Against Conventional Attacks  
*Supervisor*: Dr. Hakem Beitollahi  
*CGPA*: 3.88/4.0 (17.58/20 in Iranian Scale) **Ranked 2<sup>nd</sup>/10** in the Department; **4<sup>th</sup>/120** in the School
- Oct 2015 – Dec 2020 **B.Sc in Software Engineering**, *Azad University, South Tehran Branch*, Tehran, Iran.  
*Thesis*: Developing a Deep Neural Network for Handwritten Digit Recognition Using Python  
*CGPA (Last Two Years)*: 2.73/4.0 (14.65/20 in Iranian Scale)

### Publications

- 2025 M. Pandi, **M. Moghaddas**, H. Beitollahi. "TestLock: A Testability Logic Locking Method Against Machine Learning-Based Oracle-Less Attacks." *The Journal of Supercomputing*, 81(14), 1–25. (DOI)
- 2025 **M. Moghaddas**, M. Pandi, H. Beitollahi. "OPUF: Obfuscated PUF Against Machine Learning Modeling Attacks." *Journal of Circuits, Systems and Computers*, 34(11), 2550245. (DOI)
- 2024 H. Beitollahi, M. Pandi, **M. Moghaddas**. "Edge-Sorter: A Hardware Sorting Engine for Area & Power Constrained Edge Computing Devices." *Microprocessors and Microsystems*, 105, 105006. (DOI)

### Honors & Awards

- 2022 **2<sup>nd</sup> Place**, Dept. of Computer Systems Architecture Engineering, IUST (2<sup>nd</sup> out of 10 graduate students by GPA)
- 2022 **4<sup>th</sup> Place**, School of Computer Engineering, IUST (4<sup>th</sup> out of 120 graduate students by GPA)
- Recognized as Exceptional Talent** — Eligible for Direct Ph.D. Track, IUST
- Member**, National Elites Foundation of Iran

### Research Experience

- Fall 2021 – Present **Research Assistant**, *Dr. Beitollahi's Hardware Security Lab, IUST*, Tehran, Iran.
- Designed and evaluated novel PUF architectures with improved resilience against machine learning modeling attacks (M.Sc. thesis).
  - Contributed to the design and analysis of testability-based logic locking methods (TestLock paper).
  - Co-developed a hardware sorting engine optimized for area- and power-constrained edge devices (Edge-Sorter paper).

### Peer Review Service

- 2026 Reviewer, *Scientific Reports* (Nature Research, Q1)
- 2025 Reviewer, *The Journal of Supercomputing* (Springer, Q2)

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## Teaching Experience

### Head Teaching Assistant

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|-------------|---------------------------------|--------------------------------|
| Fall 2024   | Computer-Aided Design           | <i>Dr. H. Beitollahi, IUST</i> |
| Spring 2021 | Database Design                 | <i>Dr. M. Hassanpour</i>       |
| Spring 2021 | Research & Presentation Methods | <i>Dr. M. Hassanpour</i>       |

### Teaching Assistant

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|--------------------|--------------------------------|----------------------------------|
| Spring 2024        | Computer-Aided Design          | <i>Dr. H. Beitollahi, IUST</i>   |
| Fall 2023          | Advanced Computer Architecture | <i>Dr. P. Lotfi-Kamran, IUST</i> |
| Spring 2022        | Computer Architecture          | <i>Dr. H. Beitollahi, IUST</i>   |
| Fall 2021          | Computer-Aided Design          | <i>Dr. H. Beitollahi, IUST</i>   |
| Spring 2020 – 2021 | Compiler Design                | <i>Dr. M. Hassanpour</i>         |

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## Selected Projects

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| Spring 2026 | <b>ASCON Implementation and Side-Channel Analysis..</b><br>Ongoing implementation of ASCON with DPA, CPA, and machine-learning-based side-channel evaluation; focused on leakage assessment, power-trace analysis, attack modeling, and implementation-level security. ( <a href="#">Repository</a> ) |
| Spring 2025 | <b>Hardware Cryptography Implementations.</b><br>Complete AES implementation on Artix-7 FPGA; stream ciphers (A5/1, Trivium); Keccak hash function; hardware-efficient Kogge-Stone adder and Karatsuba multiplier architectures. ( <a href="#">Repository</a> )                                       |
| Spring 2025 | <b>Side-Channel Analysis of AES.</b><br>In-depth security evaluation of multiple AES implementations (T-Table, BRAM, pipelined, byte-serial, software) against timing attacks, cache-timing, SPA, DPA, CPA, template attacks, and Correlation-Enhanced Collision attacks.                             |
| Fall 2022   | <b>FPGA–DAC Interface (VHDL).</b><br>Design and implementation of an FPGA communicating with an AD9117 dual DAC via SDIO in VHDL. ( <a href="#">Repository</a> )                                                                                                                                      |
| Spring 2022 | <b>Fault Analysis on a Simple CPU (Verilog).</b><br>Designed a simple CPU in Verilog; performed fault injection, fault collapsing, and fault coverage analysis following Navabi's <i>Digital System Test and Testable Design</i> . ( <a href="#">Repository</a> )                                     |
| Spring 2022 | <b>Hardware Trojan Attack Replication.</b><br>Reproduced results from "Hardware Trojan Attack in Embedded Memory" (ACM DL) using Synopsys HSPICE.                                                                                                                                                     |

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## Technical Skills

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| Programming      | C, Rust, Python, VHDL/Verilog, SystemVerilog                                                                     |
| EDA & Simulation | Xilinx Vivado, ISE, Synopsys Design Compiler, Cadence, Synopsys HSPICE, Yosys, ABC, OpenROAD, JasperGold, MATLAB |
| Hardware         | Artix-7 (XC7A35-1FTG256-2), Xilinx-7 Series FPGAs, RISC-V                                                        |
| Languages        | Persian (Native), English (Professional), Arabic (Basic)                                                         |

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## References

### Reference 1 Dr. Hakem Beitollahi

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|-------------|----------------------------------------------------------------------------------|
| Role        | M.Sc. Supervisor                                                                 |
| Institution | Soran University                                                                 |
| Email       | <a href="mailto:hakem.beitollahi@soran.edu.iq">hakem.beitollahi@soran.edu.iq</a> |
| Note        | Primary Research Collaborator                                                    |

### Reference 2 Dr. Hosseini Monazzah

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|-------------|--------------------------------------------------------------|
| Role        | M.Sc. Thesis Internal Examiner                               |
| Institution | IUST                                                         |
| Email       | <a href="mailto:monazzah@iust.ac.ir">monazzah@iust.ac.ir</a> |
| Note        | Course Instructor & Examiner                                 |